PickHacks 2026: OASIS

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1 Introduction

OASIS (Optimal Aqueduct Siting Intelligence System) is a system to find optimal sites for new reservoirs for municipal water. We use an algorithm that uses water demand levels, distance, and pumping costs to choose an optimal location for a reservoir and/or a water tower.

1.1 Purpose Statement

Providing running water is one of the most important functions that a municipality is tasked with. Our goal with OASIS is to provide an example tool that shows how services could be placed on a grid of a semi-realistic city demand map.

2 Assumptions

- Constant Pipe Diameter
- Constant Price per kWh
- Level City (No Inclines/Declines)
- Tiles are 1/4 Kilometer squares

3 Front End

We used Claude Sonnet 4.6 to generate a Front end for our project, which displays a random heatmap representing demand in an area. It also includes sliders to change key variables in the program.

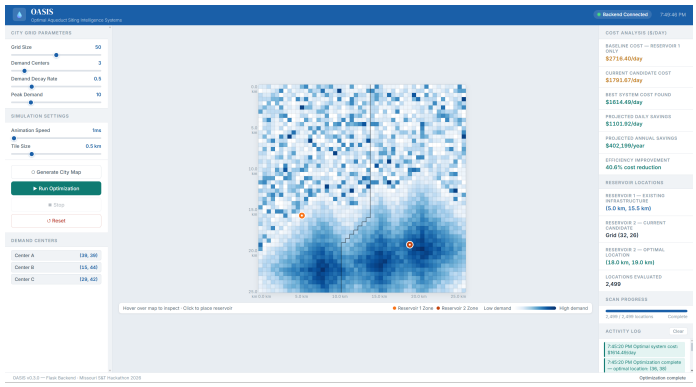

Figure 1: Front end

4 Backend

The Backend handles the calculations required to find the optimal location. It also handles data representation. The bulk of our time was used on the math specifically, however, we cannot be certain in our calculations regarding the fiscal cost per day. We were not able to find any publicly available cost data to compare our calculations to. An extension to the project may be to do research regarding more real-life examples (actual city data + map overlay). A different, more efficient search algorithm would probably also a net benefit.

5 Results

We successfully created a system that finds the optimal location of a second reservoir, and used Claude to quickly build us a front end using flask. See the rest of this repository, and follow to readme to run the program for yourself!

5.1 Major Takeaways

- Water management is complicated! During our research, we discovered that there is a lot of variables that go into what we are trying to do here! If there was more time to do research, we would probably consult a Civil Engineer to find out what we should take into consideration.
- We learned our limits, during the course of working on our project, we put a lot of workload onto ourselves. We are capable of a lot, but we had to make tough decisions on the scope of this project in order to finish in time.
- We had fun! While we definitely worked ourselves as much as possible, we also intentionally took time to allow our minds to rest so that we could still be effective when we came back to work on the project.